Chapter 11

Feasible Direction Methods

11.1 Feasible Point Methods

A feasible point method requires that all iterate points x_k generated are feasible points of the constraints. For general constrained optimization problems (8.1.1)-(8.1.3), given a current iterate $x_k \in X$, if we can find a descent direction d which is also a feasible direction at x_k, namely

$$d^T \nabla f(x_k) < 0, \tag{11.1.1}$$

$$d \in \text{FD } (x_k, X), \tag{11.1.2}$$

there must exist new feasible points in the form of $x_k + \alpha d$ with the property that $f(x_k + \alpha d) < f(x_k)$. Here FD (x_k, X) is defined by Definition 8.2.1. A direction d satisfying (11.1.1)-(11.1.2) is called a feasible descent direction at x_k.

Let $c_1 \in (0, 1)$ be a given positive constant, x_k be any point in the feasible set X, and d be a vector that satisfies (11.1.1)-(11.1.2). We call α a feasible point Armijo step along direction d at point x_k if $\alpha > 0$ satisfies

$$f(x_k + \alpha d) \leq f(x_k) + \alpha c_1 d^T \nabla f(x_k), \tag{11.1.3}$$

and

$$f(x_k + 2\alpha d) > f(x_k) + 2\alpha c_1 d^T \nabla f(x_k) \tag{11.1.4}$$

holds when $x_k + 2\alpha d \in X$.

Lemma 11.1.1 *Assume that $x_k \in X$ and d satisfies (11.1.1)-(11.1.2). Let α be a feasible point Armijo step along direction d at point x_k, then*

$$f(x_k + \alpha d) \leq f(x_k) - \frac{c_1(1-c_1)}{M}\left[\frac{d^T \nabla f(x_k)}{\|d\|_2}\right]^2 \tag{11.1.5}$$

if $x_k + 2\alpha d \in X$, and

$$f(x_k + \alpha d) \leq f(x_k) + c_1 \frac{\Gamma(x_k)}{2\|d\|_2} d^T \nabla f(x_k), \tag{11.1.6}$$

if $x_k + 2\alpha d \notin X$, where $M = \max_{0 \leq t \leq 2} \|\nabla^2 f(x_k + td)\|_2$ and $\Gamma(\bar{x})$ is the distance from $\bar{x}$ to the set of all infeasible points, namely

$$\Gamma(\bar{x}) = \inf_{y \notin X} \|\bar{x} - y\|. \tag{11.1.7}$$

Proof. First assume that $x_k + 2\alpha d \in X$. It follows from (11.1.4) and Taylor expansion that

$$\begin{aligned} 2\alpha c_1 d^T \nabla f(x_k) &< 2\alpha d^T \nabla f(x_k) + \frac{1}{2}(2\alpha d)^T \nabla^2 f(x_k + \eta_k 2\alpha d)(2\alpha d) \\ &\leq 2\alpha d^T \nabla f(x_k) + 2\alpha^2 M \|d\|_2^2, \end{aligned} \tag{11.1.8}$$

where $\eta_k \in (0,1)$. From the above inequality we can obtain that

$$\alpha > -\frac{(1-c_1)}{M\|d\|_2^2} d^T \nabla f(x_k). \tag{11.1.9}$$

Inequality (11.1.15) follows from the above relation and (11.1.3).

Now we consider the case when $x_k + 2\alpha d \notin X$. It follows from the definition of $\Gamma(x)$ that

$$2\alpha \|d\|_2 > \Gamma(x_k). \tag{11.1.10}$$

Thus, $\alpha > \frac{\Gamma(x_k)}{2\|d\|_2}$. This inequality and condition (11.1.3) imply (11.1.6). $\square$

The algorithm given below is a simple algorithm for calculating a feasible point Armijo step. It tries to find an acceptable step by repeatedly doubling or halving the step.

Algorithm 11.1.2

Step 1. *Given* $x \in X$, $d \in DF(x, X)$ *and* $d^T \nabla f(x) < 0$;
given $c_1 \in (0, 1)$;
let $\alpha_{\max} = +\infty$, $\alpha = 1$.

Step 2. *if*

$$f(x + \alpha d) > f(x) + c_1 \alpha d^T \nabla f(x),$$

or $x + \alpha d \notin X$ *go to Step 3;*
if $\alpha_{\max} < +\infty$ *then stop;*
$\alpha := 2\alpha$; *Go to Step 2.*

Step 3. $\alpha_{\max} := \alpha$; $\alpha := \alpha/2$; *Go to Step 2.* □

It is easy to see that Algorithm 11.1.2 terminates after finitely many iterations with a feasible point Armijo step unless $x + 2^k d \in X$ for all k and $f(x + 2^k d) \to -\infty$. Instead of simply doubling or halving the trial step, we can also use quadratic or cubic interpolations in the above algorithm to accelerate the convergence speed.

For any $x \in X$ and $d \in \mathrm{FD}\ (x, X)$, we call the step $\alpha^* > 0$ that satisfies

$$\alpha^* : \min_{\substack{\alpha > 0 \\ x + \alpha d \in X}} f(x + \alpha d) \tag{11.1.11}$$

a feasible point exact line search step.

Lemma 11.1.3 *Assume that* $x \in X$, $d \in FD\ (x, X)$, *and* α^* *satisfies (11.1.11). It follows that*

$$f(x) - f(x + \alpha^* d) \geq \frac{1}{2M} \left[\frac{d^T \nabla f(x)}{\|d\|_2} \right]^2, \tag{11.1.12}$$

or

$$f(x) - f(x + \alpha^* d) \geq -\frac{\Gamma(x)}{2\|d\|_2} d^T \nabla f(x), \tag{11.1.13}$$

where $M = \max_{t \geq 0} \|\nabla^2 f(x + td)\|_2$ *and where* $\Gamma(x)$ *is defined by (11.1.7).*

Proof. From Taylor expansion, it follows that

$$f(x + \alpha d) \leq f(x) + \alpha d^T \nabla f(x) + \frac{M}{2} \|d\|_2^2 \alpha^2 \triangleq \phi(\alpha). \tag{11.1.14}$$

Let $\alpha_0 = -d^T \nabla f(x) / (M \|d\|_2^2)$. If $x + \alpha_0 d \in X$, we have that

$$\begin{aligned} f(x+\alpha^* d) &\leq f(x+\alpha_0 d) \leq \phi(\alpha_0) \\ &= f(x) - \frac{1}{2M}\left[\frac{d^T \nabla f(x)}{\|d\|_2}\right]^2. \end{aligned} \tag{11.1.15}$$

If $x+\alpha_0 d \notin X$, it follows that $\alpha_0 \geq \Gamma(x)/\|d\|$. From the convexity of $\phi(\alpha)$, we can show that

$$\begin{aligned} f(x) - f(x+\alpha^* d) &\geq \sup_{0<\alpha<\Gamma(x)/\|d\|_2} [f(x) - f(x+\alpha d)] \\ &\geq \sup_{0<\alpha<\Gamma(x)/\|d\|_2} [f(x) - \phi(\alpha)] \\ &= f(x) - \phi[\Gamma(x)/\|d\|_2] \\ &\geq \frac{\Gamma(x)}{\|d\|_2 \alpha_0}[f(x) - \phi(\alpha_0)] \\ &= \frac{-\Gamma(x)}{2\|d\|_2} d^T \nabla f(x). \end{aligned} \tag{11.1.16}$$

The above two inequalities indicate that the lemma is true. □

Having the technique of searching along a feasible direction in the feasible region, we can solve a constrained optimization problem iteratively as long as we can find a feasible descent direction in every iteration. However, it is not always possible to find a feasible descent direction. For example, for the constraint

$$c(x,y) = y - x^2 = 0, \quad \begin{pmatrix} x \\ y \end{pmatrix} \in \Re^2, \tag{11.1.17}$$

$\mathrm{FD}((x,y),X) = \mathit{\Phi}$ at every feasible point. Therefore no feasible descent direction exists at any feasible point. Fortunately, when the feasible set X is convex, at any point $x \in X$ there exists a feasible descent direction provided that x is not a KKT point. We write this result in the form of a lemma as follows.

Lemma 11.1.4 *Assume that $x \in X$, X is a convex set and $f(x)$ is a convex function. Then there exists a feasible descent direction at x if and only if x is not a minimizer of problem (8.1.1)-(8.1.3).*

Proof. It is obvious that there exists no feasible descent direction at x if x is a minimizer.

Now assume that x is not a minimizer, then there exists an $\hat{x} \in X$ such that

$$f(\hat{x}) < f(x). \tag{11.1.18}$$

Because $f(x)$ is a convex function, it follows from (11.1.18) that

$$d^T \nabla f(x) < 0,$$

where $d = \hat{x} - x$. Because of the convexity of X, $d \in$ FD (x, X). Therefore d is a feasible descent direction. □

A general algorithm that uses feasible descent directions is given as follows.

Algorithm 11.1.5

Step 1. *Given initial point* $x_1 \in X$, $k := 1$;

Step 2. *If no vector* d *satisfies (11.1.1)-(11.1.2) then stop;*
find d_k *that satisfies (11.1.1)-(11.1.2);*

Step 3. *Carry out a certain feasible point search, obtaining* $\alpha_k > 0$.

Step 4. $x_{k+1} = x_k + \alpha_k d_k$; $k := k + 1$; *Go to Step 2.*

We can use a feasible point exact line search or a feasible point Armijo search to obtain α_k in Step 3 of the above algorithm.

From example (11.1.17), even if Algorithm 11.1.5 terminates, it may not stop at a stationary point. However, when the objective function $f(x)$ is convex and when the feasible set is convex, x_k must be the optimal solution if Algorithm 11.1.5 terminates at iteration k.

An important issue is the choice of d_k that satisfies (11.1.1)-(11.1.2). Consider the very special case when $X = \Re^n$. Let $f(x)$ be a uniformly convex function defined on $\Re^n$. Assume that d_k is the search direction at the k-th iteration satisfying

$$d_k^T \nabla f(x_k) < 0. \tag{11.1.19}$$

Let θ_k be the angle between d_k and the steepest descent direction $-\nabla f(x_k)$, namely

$$\cos \theta_k = -\frac{d_k^T \nabla f(x_k)}{\|d_k\|_2 \|\nabla f(x_k)\|}. \tag{11.1.20}$$

Lemma 11.1.6 *For an unconstrained optimization problem, assume that the objective function is twice continuously differentiable and uniformly convex and that a line search algorithm with* $x_{k+1} = x_k + \alpha_k d_k$ *and* $\|\nabla f(x_k)\| \neq 0$ *for all* k*, satisfies*

$$\sum_{k=1}^{\infty} \cos^2 \theta_k < +\infty, \tag{11.1.21}$$

where $\cos\theta_k$ *is defined by (11.1.20). Then*

$$\lim_{k\to\infty} \inf \|\nabla f(x_k)\| > 0. \tag{11.1.22}$$

Proof. Because $f(x)$ is uniformly convex, there exists x^* such that

$$f(x^*) = \min_{x\in\Re^n} f(x). \tag{11.1.23}$$

It is obvious that (11.1.22) is equivalent to

$$\lim_{k\to\infty} f(x_k) > f(x^*). \tag{11.1.24}$$

Define $X_1 = \{x | f(x) \le f(x_1)\}$ and

$$m_1 = \min_{x\in X_1} \min_{\|d\|_2=1} d^T \nabla^2 f(x) d, \tag{11.1.25}$$

$$M_1 = \max_{x\in X_1} \max_{\|d\|_2=1} d^T \nabla^2 f(x) d. \tag{11.1.26}$$

It can be shown that $0 < m_1 \le M_1 < +\infty$ because $f(x)$ is uniformly convex. Therefore,

$$\begin{aligned} f(x_k) - f(x_{k+1}) &\le f(x_k) - \min_{t>0} f(x_k + t d_k) \\ &\le \frac{1}{2m_1} \|\nabla f(x_k)\|_2^2 \cos^2 \theta_k \\ &\le \frac{\cos^2 \theta_k}{2m_1} (M_1 \|x_k - x^*\|_2)^2 \\ &\le \frac{\cos^2 \theta_k}{2} \left(\frac{M_1}{m_1}\right)^2 [f(x_k) - f(x^*)]. \end{aligned} \tag{11.1.27}$$

Consequently, it follows that

$$f(x_{k+1}) - f(x^*) \ge \left(1 - \frac{M_1^2}{2m_1^2} \cos^2 \theta_k\right) [f(x_k) - f(x^*)], \tag{11.1.28}$$

for all k. Assumption (11.1.21) implies the existence of k_0 such that

$$\frac{M_1^2}{2m_1^2}\cos^2\theta_k < 1, \quad \forall k \geq k_0. \tag{11.1.29}$$

Because $\|\nabla f(x_{k_0})\| \neq 0$, we have that $f(x_{k_0}) - f(x^*) = \delta > 0$. From (11.1.21) there exists $\eta > 0$ such that

$$\textstyle\prod_{j=k_0}^{\infty}\left(1 - \frac{M_1^2}{2m_1^2}\cos^2\theta_k\right) \geq \eta > 0. \tag{11.1.30}$$

Thus, it follows from (11.1.28) and (11.1.30) that

$$f(x_k) - f(x^*) \geq \eta\delta > 0$$

for all $k \geq k_0$. This implies that (11.1.24). $\square$

The above lemma tells us that we require

$$\sum_{k=1}^{\infty}\cos^2\theta_k = +\infty, \tag{11.1.31}$$

to ensure the algorithm converging to a stationary point.

Similar to the steepest descent direction, we can define the feasible steepest descent direction as follows.

Definition 11.1.7 *Let $x \in X$; if a vector d in the closure of FD(x, X) solves*

$$\min_{\substack{d\in \mathrm{FD}\ (x,X)\\ d\neq 0}} \frac{d^T\nabla f(x)}{\|d\|_2}, \tag{11.1.32}$$

it is called a feasible steepest descent direction.

Because $\mathrm{FD}(x, X)$ may not be a closed set, the minimum of (11.1.32) can not be reached by any $d \in \mathrm{FD}\ (x, X)$. Thus, a feasible steepest descent direction may not belong to $\mathrm{FD}(x, X)$. Therefore, it is not easy to generate the steepest direction directly to constrained optimization by simply making the steepest descent direction "feasible".

Consider the inequality constrained optimization problem

$$\min f(x), \tag{11.1.33}$$

$$\text{s.t. } c_i(x) \geq 0 \quad i = 1, \cdots, m. \tag{11.1.34}$$

Let $x_k \in X$. It is obvious that $I(x_k) = \{i | c_i(x_k) = 0\}$. In order to find a feasible descent direction at the k-th iteration, we consider the approximate subproblem

$$\min \ \alpha d^T \nabla f(x_k), \tag{11.1.35}$$

$$\text{s.t. } x_k + \alpha d \in X. \tag{11.1.36}$$

As the aim for constructing this subproblem is to find a search direction, we can assume that $\|\alpha d\|$ is very small. When $\|\alpha d\|$ is sufficiently small, (11.1.36) is equivalent to

$$c_j(x_k + \alpha d) \geq 0, \quad j \in I(x_k). \tag{11.1.37}$$

The above inequalities hold if we require that

$$\alpha d^T \nabla c_j(x_k) - \frac{1}{2} M \alpha^2 \|d\|_2^2 \geq 0, \quad j \in I(x_k), \tag{11.1.38}$$

where $M > 0$ is an upper bound for

$$\max_{x \in X} \max_{j \in I(x_k)} \|\nabla^2 c_j(x)\|_2. \tag{11.1.39}$$

Replacing αd by d, we can obtain the following subproblem

$$\min d^T \nabla f(x_k), \tag{11.1.40}$$

$$\text{s.t. } d^T \nabla c_i(x_k) - \frac{M}{2} \|d\|_2^2 \geq 0, \quad i \in I(x_k). \tag{11.1.41}$$

By further replacing d by Md, the above problem can be rewritten as

$$\min d^T \nabla f(x_k), \tag{11.1.42}$$

$$\text{s.t. } d^T \nabla c_i(x_k) - \frac{1}{2} \|d\|_2^2 \geq 0, \quad i \in I(x_k). \tag{11.1.43}$$

The dual problem for the above problem is

$$\max_{\lambda} \min_{d \in \Re^n} \left[d^T \nabla f(x_k) - \sum_{i \in I(x_k)} \lambda_i \left(d^T \nabla c_i(x_k) - \frac{1}{2} \|d\|_2^2 \right) \right] \tag{11.1.44}$$

$$\text{s.t. } \lambda_i \geq 0, \quad i \in I(x_k). \tag{11.1.45}$$

The above problem can be written in the equivalent form

$$\min_{\lambda} \quad \frac{\left\|\nabla f(x_k) - \sum_{i\in I(x_k)} \lambda_i \nabla c_i(x)\right\|_2^2}{\sum_{i\in I(x_k)} \lambda_i}, \tag{11.1.46}$$

$$\text{s.t.} \quad \lambda_i \geq 0, \quad i \in I(x_k), \qquad \sum_{i\in I(x_k)} \lambda_i > 0. \tag{11.1.47}$$

Define

$$d(\lambda) = -\frac{1}{\sum_{i\in I(x_k)} \lambda_i} \left(\nabla f(x_k) - \sum_{i\in I(x_k)} \lambda_i \nabla c_i(x_k) \right). \tag{11.1.48}$$

The objective function in (11.1.46) can be written as

$$\phi(\lambda) = \sum_{i\in I(x_k)} \lambda_i \|d(\lambda)\|_2^2. \tag{11.1.49}$$

Direct calculations show that

$$\nabla\phi(\lambda) = \begin{bmatrix} 2d(\lambda)^T \nabla c_{k_1}(x_k) - \|d(\lambda)\|_2^2 \\ \vdots \\ 2d(\lambda)^T \nabla c_{k_I}(x_k) - \|d(\lambda)\|_2^2 \end{bmatrix}, \tag{11.1.50}$$

where $\{k_1, k_2, \cdots, k_I\}$ are the elements of $I(x_k)$. Furthermore, we have that

$$\nabla^2\phi(\lambda) = \frac{2}{\sum_{i\in I(x_k)} \lambda_i} T(\lambda)^T T(\lambda), \tag{11.1.51}$$

where

$$T(\lambda) = (d(\lambda) - \nabla c_{k_1}(x_k), d(\lambda) - \nabla c_{k_2}(x_k), \cdots, d(\lambda) - \nabla c_{k_I}(x_k)). \tag{11.1.52}$$

Thus, $\phi(\lambda)$ is a convex function. Let $\lambda^{(k)}$ be a solution of (11.1.46)-(11.1.47), then $d(\lambda^{(k)})$ is a solution of problem (11.1.40)-(11.1.41). In that case, x_k is a KKT point of (11.1.33)-(11.1.34) if $d(\lambda^{(k)}) = 0$, and $d(\lambda^{(k)})$ is a feasible descent direction at x_k satisfying

$$d(\lambda^{(k)})^T \nabla f(x_k) = \frac{\|d(\lambda^{(k)})\|_2^2}{\sum_{i\in I(x_k)} \lambda_i^{(k)}} < 0, \tag{11.1.53}$$

if $d(\lambda^{(k)}) \neq 0$.

It is not difficult to show that subproblem (11.1.40)-(11.1.41) has nonzero minimum if and only if

$$\begin{aligned} d^T\nabla f(x_k) &< 0, && (11.1.54)\\ d^T\nabla c_i(x_k) &> 0, \quad i \in I(x_k) && (11.1.55)\end{aligned}$$

has a solution. That is to say, when (11.1.54)-(11.1.55) has a solution we can obtain a feasible descent direction by solving (11.1.40)-(11.1.41). On the other hand, if (11.1.54)-(11.1.55) has no solution, similar to Lemma 8.2.5 it can be shown that there exist $\lambda_i^*(i \in I(x_k)) \geq 0$ and $\lambda_0^* \geq 0$ such that

$$\lambda_0^*\nabla f(x_k) - \sum_{i\in I(x_k)} \lambda_i^*\nabla c_i(x^*) = 0, \tag{11.1.56}$$

and that $\sum_{i\in I(x_k)} \lambda_i^{*2} + \lambda_0^* \neq 0$. Therefore we know that x_k is a Fritz John point of the original optimization problem (11.1.33)-(11.1.34).

Another subproblem for finding a feasible descent direction is directly based on (11.1.54)-(11.1.55), having the form:

$$\begin{aligned} &\min\ \sigma && (11.1.57)\\ \text{s.t. } d^T\nabla f(x_k) &\leq +\sigma, && (11.1.58)\\ d^T\nabla c_i(x_k) &\geq -\sigma, \quad i \in I(x_k), && (11.1.59)\\ \|d\| &\leq 1. && (11.1.60)\end{aligned}$$

It is easy to see that the minimum of the above subproblem $\sigma^* = 0$ if and only if (11.1.54)-(11.1.55) has no solutions.

11.2 Generalized Elimination

Consider the equality constrained problem

$$\begin{aligned} \min \quad & f(x) && (11.2.1)\\ \text{s.t.} \quad & c(x) = 0, && (11.2.2)\end{aligned}$$

where $c(x) = (c_1(x), \cdots, c_m(x))^T$. Assume that we have a certain partition on the variable x:

$$x = \begin{bmatrix} x_B \\ x_N \end{bmatrix}, \tag{11.2.3}$$

where $x_B \in \Re^m$, $x_N \in \Re^{n-m}$. Therefore (11.2.2) can be written as

$$c(x_B, x_N) = 0. \tag{11.2.4}$$

Suppose that we can solve x_B from (11.2.4), namely

$$x_B = \phi(x_N), \tag{11.2.5}$$

then (11.2.1)-(11.2.2) is equivalent to

$$\min_{x_N \in \Re^{n-m}} f(x_B, x_N) = f(\phi(x_N), x_N) = \tilde{f}(x_N). \tag{11.2.6}$$

The vector

$$\tilde{g}(x_N) = \nabla x_N \tilde{f}(x_N) \tag{11.2.7}$$

is called the reduced gradient. It is easy to verify that

$$\tilde{g}(x_N) = \frac{\partial}{\partial x_N} f(x_B, x_N) + \frac{\partial x_B^T}{\partial x_N} \frac{\partial}{\partial x_B} f(x_B, x_N). \tag{11.2.8}$$

From (11.2.4) we can see that $\frac{\partial x_B^T}{\partial x_N}$ satisfies that

$$\frac{\partial x_B^T}{\partial x_N} \frac{\partial}{\partial x_B} c(x_B, x_N)^T + \frac{\partial}{\partial x_N} c(x_B, x_N)^T = 0. \tag{11.2.9}$$

If $\frac{\partial c^T}{\partial x_B}$ is nonsingular, the above two equations imply that

$$\begin{aligned} \tilde{g}(x_N) \quad = \quad & \frac{\partial f(x_B, x_N)}{\partial x_N} \\ & - \frac{\partial c(x_B, x_N)^T}{\partial x_N} \left[\frac{\partial c(x_B, x_N)^T}{\partial x_B} \right]^{-1} \frac{\partial f(x_B, x_N)}{\partial x_B}. \end{aligned} \tag{11.2.10}$$

Therefore, the reduced gradient can be expressed as the gradient of the Lagrangian function at the reduced space:

$$\tilde{g}(x_N) = \frac{\partial}{\partial x_N} [f(x) - \lambda^T c(x)], \tag{11.2.11}$$

where λ is a multiplier satisfying

$$\frac{\partial f(x)}{\partial x_B} = \frac{\partial c^T(x)}{\partial x_B} \lambda. \tag{11.2.12}$$

In other words, when the Lagrange multiplier λ is chosen as

$$\left[\frac{\partial c^T(x_B)}{\partial x_B}\right]^{-1}\frac{\partial f(x)}{\partial x_B}, \tag{11.2.13}$$

we have that

$$\nabla_x L(x,\lambda) = \begin{bmatrix} 0 \\ \tilde{g}(x_N) \end{bmatrix}. \tag{11.2.14}$$

Therefore, the reduced gradient can be viewed as the nonzero part of the gradient of the Lagrangian function.

Using the reduced gradients, we can construct line search directions for the unconstrained problem (11.2.6). For example, we can use the steepest descent direction

$$\bar{d}_k = -\tilde{g}((x_N)_k) \tag{11.2.15}$$

or the quasi-Newton direction

$$\bar{d}_k = -B_k^{-1}\tilde{g}((x_N)_k). \tag{11.2.16}$$

Here the subscript k indicates the iterate number, B_k is an approximate Hessian matrix which can be updated from iteration to iteration(for example, by BFGS formula). It is worth pointing out that carrying out a line search

$$\min_{\alpha\geq 0} f(\phi((x_N)_k+\alpha\bar{d}_k), \quad (x_N)_k+\alpha\bar{d}_k) \tag{11.2.17}$$

on the unconstrained problem (11.2.6) is equivalent to carrying out a curve search on the original objective function $f(x)$ along the following curve:

$$c(x_B, (x_N)_k+\alpha\bar{d}_k) = 0. \tag{11.2.18}$$

Because the function $\phi(x)$ is not known explicitly, for every trial α we need to solve (11.2.18) to obtain

$$x_B = \phi((x_N)_k+\alpha\bar{d}_k) \tag{11.2.19}$$

when carrying out line searches (11.2.17). This can be done by an approximate Newton's method, namely

$$x_B^{(0)} = (x_B)_k, \tag{11.2.20}$$

$$x_B^{(i+1)} = x_B^{(i)} - \left[\frac{\partial c(x_k)^T}{\partial x_B}\right]^{-1} c(x_B^{(i)}, (x_N)_k+\alpha\bar{d}_k). \tag{11.2.21}$$

Because Newton's method converges quadratically, usually an acceptable x_B will be obtained after applying (11.2.21) for a few iterations. If $x_B^{(i)}$ does not converge after some iterations, α should be reduced to continue the line search procedure.

The following is a general framework of the variable elimination method.

Algorithm 11.2.1 *(Variable Elimination Method)*

Step 1. *Given a feasible point* $x_1 \in X$, $\epsilon \geq 0$, $k = 1$;

Step 2. *Compute*

$$\frac{\partial c(x_k)^T}{\partial x} = \begin{bmatrix} A_B \\ A_N \end{bmatrix}, \tag{11.2.22}$$

where the partition satisfies that A_B *is nonsingular.* *Compute* λ *by (11.2.12), and* $\tilde{g}_k$ *by (11.2.11).*

Step 3. *If* $\|\tilde{g}_k\| \leq \epsilon$ *then Stop;* *Generate a feasible descent direction* $\bar{d}_k$ *satisfying*

$$\bar{d}_k^T \tilde{g}_k < 0. \tag{11.2.23}$$

Step 4. *Carry out line search (11.2.17) obtaining* $\alpha_k > 0$, *Let* $x_{k+1} = (\phi((x_N)_k + \alpha_k \bar{d}_k), (x_N)_k + \alpha_k \bar{d}_k)$, $k := k+1$; *go to Step 2.*

It is very easy to see that the above algorithm is in fact a descent method for the unconstrained optimization problem (11.2.6). The only thing that we should keep in mind is that the partition (x_B, x_N) may differ from iteration to iteration. Using the convergence results of descent methods for unconstrained optimization, we can easily establish the following result.

Theorem 11.2.2 *Assume that* $f(x)$ *and* $c(x)$ *are twice continuously differentiable. If* $[(\nabla c(x)^T)^T \ \nabla c(x)^T]^{-1}$ *is bounded above uniformly on the feasible set* X, *Algorithm 11.2.1 with exact line searches and the assumption*

$$\sum \cos^2 \langle \bar{d}_k, \tilde{g}_k \rangle = \infty \tag{11.2.24}$$

ensures that

$$\liminf_{k \to \infty} \|(\nabla f(x_k) - \nabla c(x_k)^T \lambda_k)\| = 0, \tag{11.2.25}$$

or

$$\lim_{k\to\infty} f(x_k) = -\infty, \tag{11.2.26}$$

where $\lambda_k = [\nabla c(x_k)^T]^+ \nabla f(x_k)$.

Let $\bar{d}_k = -\tilde{g}_k$, then (11.2.23) and (11.2.24) hold. In that case, Algorithm 11.2.1 is exactly the steepest descent method in the lower dimensional space using the variable partition.

Consider any nonsingular matrix $S \in \Re^{n\times n}$ and variable transformation:

$$x = Sw. \tag{11.2.27}$$

We partition the variable w:

$$w = \begin{bmatrix} w_B \\ w_N \end{bmatrix}, \tag{11.2.28}$$

where $w_B \in \Re^m$, $w_N \in \Re^{n-m}$. Using the constrained condition

$$c((S)_B w_B + (S)_N w_N) = 0 \tag{11.2.29}$$

to eliminate variable w_B, namely

$$w_B = \bar{\phi}(w_N). \tag{11.2.30}$$

In this way, the optimization problem (11.2.1)-(11.2.2) is equivalent to

$$\min_{w_N \in \Re^{n-m}} f(S_B w_B + S_N w_N) = \bar{f}(w_N). \tag{11.2.31}$$

Provided that $S_B^T \nabla C(x)^T$ is nonsingular, direct calculations give that

$$\nabla_{w_N} \bar{f}(w_N) = \bar{g}(w_N) = S_N^T[\nabla f(x) - \nabla c(x)^T \lambda], \tag{11.2.32}$$

where λ satisfies

$$S_B^T[\nabla f(x) - \nabla c(x)^T \lambda] = 0. \tag{11.2.33}$$

Thus, we have obtained an elimination method based on the variable transformations at every iterations. This method is called the generalized elimination method.

Algorithm 11.2.3 *(General Elimination Method)*

Step 1. *Given a feasible point* $x_1 \in X$, $\epsilon \geq 0$, $k = 1$;

Step 2. *Construct a nonsingular matrix* S_k, *and a partition* $S_k = [(S_k)_B\ ,\quad (S_k)_N]$ *such that* $(S_k)_B^T \nabla c(x_k)^T$ *nonsingular;* *Compute* λ *by (11.2.33) and* $\bar{g}_k$ *by (11.2.32).*

Step 3. *If* $\|\bar{g}_k\| \leq \epsilon$ *then stop;* *Generate a descent direction* $\bar{d}_k$ *satisfying*

$$\bar{d}_k^T \bar{g}_k < 0; \tag{11.2.34}$$

Step 4. *Carry out line search:*

$$\min_{\alpha>0} f((S_k)_B\bar{\phi}((w_k)_N + \alpha\bar{d}_k) + (S_k)_N[(w_k)_N + \alpha\bar{d}_k]) \tag{11.2.35}$$

obtaining $\alpha_k > 0$; *let*

$$x_{k+1} = (S_k)_B\bar{\phi}((w_k)_N + \alpha_k\bar{d}_k) + (S_k)_N[(w_k)_N + \alpha_k\bar{d}_k]; \tag{11.2.36}$$

$k := k + 1$; *go to Step 2.* $\square$

In the algorithm, w_k is a vector satisfying $x_k = S_k w_k$. Similar to the elimination method, for each trial step $\alpha > 0$, we need to compute

$$w_B = \bar{\phi}((w_k)_N + \alpha\bar{d}_k), \tag{11.2.37}$$

which can be done by applying the approximate Newton's method to the nonlinear system

$$c((S_k)_B w_B + (S_k)_N[(w_k)_N + \alpha\bar{d}_k]) = 0. \tag{11.2.38}$$

That is,

$$\begin{aligned} w_B^{(i+1)} &= w_B^{(i)} - [(\nabla c(x_k)^T)^T(S_k)_B]^{-1} c((S_k)_B w_B^{(i)} \\ &+ (S_k)_N[(w_k)_N + \alpha\bar{d}_k]), \quad i = 1, 2, \cdots. \end{aligned} \tag{11.2.39}$$

It is not difficult to see that if S_k is the unit matrix in every iteration, the generalized elimination method is exactly the original elimination method.

The variable increment $x_{k+1} - x_k$ of the generalized elimination method in every iteration is actually the sum of two parts:

$$x_{k+1} = x_k + d_k^{(1)} + d_k^{(2)}, \tag{11.2.40}$$

where

$$d_k^{(1)} = \alpha_k (S_k)_N \bar{d}_k, \tag{11.2.41}$$
$$d_k^{(2)} = (S_k)_B[\bar{\phi}((w_k)_N + \alpha_k \bar{d}_k) - (w_k)_B]. \tag{11.2.42}$$

The iteration process first obtains the step $d_k^{(1)}$, then uses the approximate Newton method along the direction $d_k^{(2)}$ to find a point in the feasible set, as shown in Figure 11.2.1.

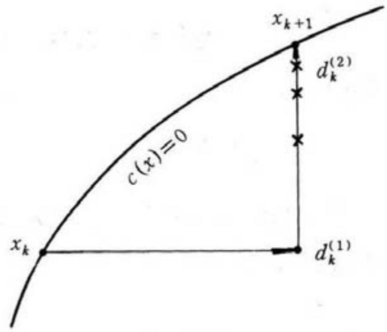

Figure 11.2.1 Iterative procedure of generalized elimination method

Looking at Figure 11.2.1, we see an undesirable property of such a process. The iteration first moves away from the feasible set, and then it comes back, though the essential idea for feasible point methods is to force all iterate points inside the feasible region. Except for very special constraints, it is unavoidable to use the technique of moving away and coming back if we require that all iterate points are feasible. But, how to make the "moving away" as small as possible? An intuitive answer is to choose $d_k^{(1)}$ to be a linearized feasible direction at x_k. It is reasonable to believe that such a $d_k^{(1)}$ would make $x_k + d_k^{(1)}$ closer to the feasible region, consequently the approximate Newton's method will bring $x_k + d_k^{(1)}$ back to the feasible region more quickly. Figure 11.2.2 illustrates the above discussions.

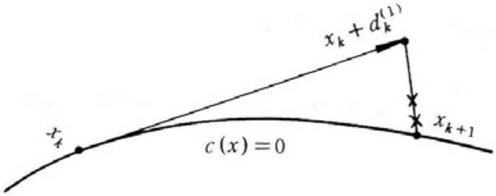

Figure 11.2.2

When $d_k^{(1)}$ is a linearized feasible direction, the method is called a feasible direction method. It is obvious that if

$$(S_k)_N^T \nabla c(x_k)^T = 0 \tag{11.2.43}$$

holds, $d_k^{(1)}$ is a linearized feasible direction. Because of this, feasible direction methods can also be viewed as special generalized elimination methods.

11.3 Generalized Reduced Gradient Method

The generalized reduced gradient method (GRG method) is in fact Algorithm 11.2.1 with $\bar{d}_k = -\tilde{g}_k$. It is the steepest descent method in the reduced space.

In each iteration, the line search can be the Armijo rule, namely reducing the trial step repeatedly until an acceptable one is obtained. The condition for accepting the new point can be the simple reduction

$$f(x_{k+1}) < f(x_k). \tag{11.3.1}$$

In each iteration, we apply (11.2.21) for at most N times to compute x_B, where N is a given positive number. If the approximate Newton's method has not converged after N iterations, we reduce the trial step α and repeat the iteration. Because the quadratic convergence of Newton's method, normally one or two iterations of (11.2.21) will return a sufficiently accurate feasible point x_{k+1}. Therefore in practice we can choose N between 3 to 6.

The algorithm is the generalized reduced gradient method with Armijo line searches requiring simple reductions.

Algorithm 11.3.1 *(Generalized Reduced Gradient Method)*

Step 1. *Given a feasible point* $x_1 \in X$, $\epsilon \geq 0$, $\bar{\epsilon} > 0$; *positive integer* M; $k := 1$.

Step 2. *Compute*

$$\nabla c(x_k)^T = \left[\begin{array}{c} A_B \\ A_N \end{array} \right], \tag{11.3.2}$$

where the partition satisfies that $A_B \in \Re^{m \times m}$ *is nonsingular; Compute* λ *from (11.2.12) and* $\tilde{g}_k$ *from (11.2.11).*

Step 3. *If* $\|\tilde{g}_k\| \leq \epsilon$ *then stop;*
let $\bar{d}_k = -\tilde{g}_k$; *and* $\alpha = \alpha_k^{(0)} > 0$.

Step 4. $x_N = (x_k)_N + \alpha \bar{d}_k$;
$x_B = (x_k)_B$; $j := 0$.

Step 5. $x_B = x_B - A_B^{-T} c(x_B, x_N)$;
compute $c(x_B, x_N)$;
if $\|c(x_B, x_N)\| \leq \bar{\epsilon}$ *then go to Step 7;*
$j := j + 1$; *if* $j < M$ *go to Step 5.*

Step 6. $\alpha := \alpha/2$, *go to Step 4.*

Step 7. *If* $f(x_B, x_N) \geq f(x_k)$ *then go to Step 6.*
$x_{k+1} = (x_B, x_N)$, $k := k + 1$; *go to Step 2.*

This algorithm is in fact a gradient method. Thus the simple reduction (11.3.1) on the objective function can not guarantee convergence. In other words, we can not show that the iterates generated by Algorithm 11.3.1 converges to a KKT point of the original optimization problem (11.2.1)-(11.2.2). There are two ways to overcome this. The first one is to use a better line search condition. For example, we can replace the simple reduction condition (11.3.1) by the Wolfe line search condition

$$\tilde{f}((x_k)_N + \alpha_k \bar{d}_k) \leq \tilde{f}((x_k)_N) + \beta \alpha_k \bar{d}_k^T \tilde{g}_k, \tag{11.3.3}$$

where α is the step length, $\beta \in (0, 1)$ is a positive constant, and $\tilde{f}(x_N)$ is defined by (11.2.6). Condition (11.3.3) can be written as

$$f(x_{k+1}) \leq f(x_k) - \alpha_k \beta \|\tilde{g}_k\|_2^2. \tag{11.3.4}$$

Thus if we replace the condition $f(x_B, x_N) \geq f(x_k)$ for rejecting a new point in Step 7 of Algorithm 11.3.1 by

$$f(x_B, x_N) > f(x_k) - \alpha\beta\|\tilde{g}_k\|_2^2, \tag{11.3.5}$$

the Wolfe line search condition (11.3.3) is satisfied. Another way is to require that the initial trial step length $\alpha_k^{(0)}$ in Step 3 of the algorithm satisfies that

$$\frac{\alpha_k^{(0)}}{\|\tilde{g}_k\|} \to 0, \tag{11.3.6}$$

$$\sum_{k=1}^{\infty} \frac{\alpha_k^{(0)}}{\|\tilde{g}_k\|} = +\infty. \tag{11.3.7}$$

Similar to convergence analyses for unconstrained optimization methods, we can prove the following convergence results.

Theorem 11.3.2 *Assume that $f(x)$, $c(x)$ are twice continuously differentiable, that the matrices A_B^{-1} in Step 2 of Algorithm 11.3.1 are bounded above uniformly, and that $\alpha_k^{(0)}$ in Step 3 satisfies that $(\alpha_k^{(0)})^{-1}$ is uniformly bounded. If the condition $f(x_B, x_N) \geq f(x_k)$ in Step 7 is replaced by (11.3.5), if $\epsilon = 0$ and if Algorithm 11.3.1 does not terminate, it follows that either*

$$\lim_{k\to\infty} \|\tilde{g}_k\| = 0 \tag{11.3.8}$$

or

$$\lim_{k\to\infty} f(x_k) = -\infty. \tag{11.3.9}$$

Theorem 11.3.3 *Assume that $f(x)$, $c(x)$ are twice continuously differentiable, that the matrices A_B^{-1} in Step 2 of Algorithm 11.3.1 are bounded above uniformly, and that $\alpha_k^{(0)}$ in Step 3 satisfies (11.3.6) and (11.3.7). Then if $\epsilon = 0$ and if Algorithm 11.3.1 does not terminate, either (11.3.8) or (11.3.9) holds.*

One advantage of the generalized reduced gradient method is the dimension of the problem is reduced due to variable elimination. The method can also make good use of the special structure of the problem such as sparsity and constant coefficients so that λ and $\tilde{g}$ can be computed quickly. Similarly, if A_B is sparse, sparse linear system solvers can be used when using the approximate Newton's method to obtain x_B. Therefore, for large-scale nonlinear programming problems there are many linear constraints or with sparse structures, the generalized reduced gradient method is one of the most efficient methods.

Because one or more systems of nonlinear equations have to be solved in the generalized reduced gradient method, its computation cost is very high if the matrix A_B is not sparse and without special structures.

11.4 Projected Gradient Method

From the discussions at the end of Section 11.2, in order to choose $d_k^{(1)}$ as a linearized feasible direction in the generalized elimination method, S_k should satisfy

$$(S_k)_N^T \nabla c(x_k)^T = 0. \tag{11.4.1}$$

Consider the case that the steepest descent direction is used in the generalized elimination method, namely

$$\bar{d}_k = -\bar{g}_k. \tag{11.4.2}$$

From (11.2.41) it follows that

$$d_k^{(1)} = -\alpha_k (S_k)_N (S_k)_N^T \nabla f(x_k). \tag{11.4.3}$$

Obviously, $(S_k)_N (S_k)_N^T$ is a linear projection from $\Re^n$ to the subspace spanned by the columns of $(S_k)_N$. Suppose $A_k = \nabla c(x_k)^T$ is full column rank, the subspace spanned by the columns of $(S_k)_N$ is the null space of A_k^T. Therefore the direction defined by (11.4.3) is actually the projection of the negative gradient of the objective function to the null space of the Jacobi matrix. If S_k satisfies

$$(S_k)_N^T (S_k)_N = I, \tag{11.4.4}$$

$(S_k)_N (S_k)_N^T$ is an orthogonal projector, and

$$\begin{aligned} P_k &= (S_k)_N (S_k)_N^T \\ &= I - A_k (A_k^T A_k)^{-1} A_k^T, \end{aligned} \tag{11.4.5}$$

when A_k has full column rank. In this case, $P_k \nabla f(x_k)$ is an orthogonal projection of $\nabla f(x_k)$ to the null space of A_k^T. Therefore, the generalized elimination method is a projected gradient method. In a practical implementation of the projected gradient method, we can use the QR factorization of A_k:

$$A_k = [Y_k \ Z_k] \begin{bmatrix} R_k \\ 0 \end{bmatrix}. \tag{11.4.6}$$

It is easy to see that we can let $(S_k)_N = Z_k$. Hence,

$$\bar{g}_k = Z_k^T g_k \tag{11.4.7}$$

is the reduced gradient and

$$d_k = -Z_k \bar{g}_k = -Z_k Z_k^T g_k \tag{11.4.8}$$

is a projection of the negative gradient to the null space of A_k^T, which is a descent direction of $f(x)$. Thus, we can choose α_k such that

$$f(x_k + \alpha_k d_k) < f(x_k). \tag{11.4.9}$$

The point $x_k + \alpha_k d_k$ may be infeasible. A feasible point can be obtained by the approximate Newton's method

$$\begin{aligned} x_k^{(1)} &= x_k + \alpha_k d_k, & (11.4.10)\\ x_k^{(i+1)} &= x_k^{(i)} - Y_k R_k^{-1} c(x_k^{(i)}), \quad i = 1, 2, \ldots. & (11.4.11) \end{aligned}$$

When $c(x_k^{i+1})$ is sufficiently small, we terminate (11.4.11) and set $x_{k+1} = x_k^{(i+1)}$. The above iteration process is essentially (11.2.39) with $(S_k)_B = Y_k$. If α_k is sufficiently small, we have that

$$\|x_{k+1} - (x_k + \alpha_k d_k)\| = O(\alpha_k^2), \tag{11.4.12}$$

therefore there exists $\alpha_k > 0$ such that

$$f(x_{k+1}) < f(x_k). \tag{11.4.13}$$

The algorithm given below is the projected gradient method with Armijo line searches requiring simple reductions.

Algorithm 11.4.1 *(Projected Gradient Method)*

Step 1. Given a feasible point $x_1 \in X$, $\epsilon \geq 0$, $\bar{\epsilon} > 0$, *a positive integer* N, $k := 1$.

Step 2. Compute the QR Factorization

$$\nabla C(x_k)^T = [Y_k \ Z_k] \begin{bmatrix} R_k \\ 0 \end{bmatrix};$$

$\bar{g}_k = Z_k^T \nabla f(x_k)$;
if $\|\bar{g}_k\| \le \epsilon$ *then stop;*
$d_k = -Z_k \bar{g}_k$; *set* $\alpha = \alpha_k^{(0)} > 0$.

Step 3. $y := x_k + \alpha d_k$; $i := 0$.

Step 4. $y := y - Y_k R_k^{-1} c(y)$;
if $\|C(y)\| \le \bar{\epsilon}$ *and* $f(y) < f(x_k)$ *then go to Step 5;*
$i := i + 1$; *if* $i < N$ *then go to Step 4;*
$\alpha = \alpha/2$; *go to Step 3.*

Step 5. $x_{k+1} := y$, $k := k + 1$; *go to Step 2.*

Similar to the generalized reduced gradient method, Algorithm 11.4.1 needs to modify its line search conditions or to impose certain conditions on the initial steplength in order to guarantee convergence.

For inequality constraints, active set technique can be used to obtain feasible directions. However, one difficulty of the active set technique is the zigzagging phenomenon, which was pointed out by Wolfe [350]. There are many ways to overcome zigzagging in feasible direction methods. The main idea for avoiding zigzagging is not to delete constraints from the active set unless it is absolutely needed.

If the search direction $d_k = -Z_k \bar{g}_k$ in the last line of Step 2 in Algorithm 11.4.1 is replaced by

$$d_k = -Z_k z_k, \tag{11.4.14}$$

where $z_k \in \Re^{n-m}$ is any vector that satisfies

$$z_k^T \bar{g}_k < 0, \tag{11.4.15}$$

then the algorithm is a general form of the linearized feasible direction method, often called feasible direction method.

Based on our definitions, the search directions in a feasible direction method is only a linearized feasible direction instead of a feasible direction. An exception is the case when all the constraints are linear functions. In this case, the linearized feasible directions are also feasible directions. Because feasible direction methods were first used for linearly constrained problems, when they are generalized to nonlinear constraints they are still called feasible direction methods. To be precise, this method, when applied to nonlinearly constrained problems, should be called linearized feasible direction method instead of feasible direction method.

For nonlinear constraints, normally linearized feasible directions are not feasible directions. Therefore searching along a linearized feasible direction may return an infeasible point. That is why Newton's method or approximate Newton's method should be applied to bring the iterate back to the feasible region before continuing the next search along a search direction and another moving back to the feasible region. This procedure leads to the sawtooth phenomenon, as indicated by Figure 11.4.1

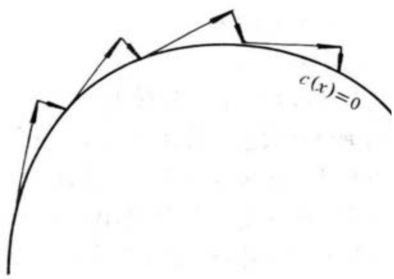

Figure 11.4.1

11.5 Linearly Constrained Problems

Feasible direction methods are very efficient for linearly constrained problems, for example, for equality constrained problem

$$\min_{x\in\Re^n} \quad f(x), \tag{11.5.1}$$

$$\text{s.t.} \qquad A^T x = b, \tag{11.5.2}$$

where $b \in \Re^m$, $A \in \Re^{n\times m}$, $rank(A) = m$, and $f(x)$ is a nonlinear function. The search direction of a feasible direction method can be expressed by

$$d_k = Z\bar{d}_k, \tag{11.5.3}$$

where $\bar{d}_k \in \Re^{n-m}$, and $Z \in \Re^{n\times(n-m)}$ satisfying

$$\begin{aligned} A^T Z &= 0, && (11.5.4)\\ \bar{d}_k^T Z^T \nabla f(x_k) &< 0. && (11.5.5) \end{aligned}$$

Specifically, we can let $\bar{d}_k = -Z^T\nabla f(x_k)$, which leads to the following feasible direction method based on steepest descent direction.

Algorithm 11.5.1

Step 1. *Given a feasible point* x_1;
Compute Z *such that* $A^T Z = 0$ *and* $Rank(Z) = n - m$;
$k = 1$, $\epsilon \geq 0$.

Step 2. $d_k = -ZZ^T\nabla f(x_k)$; *if* $\|d_k\| \leq \epsilon$ *then stop;*
Carry out line search along d_k *obtaining* $\alpha_k > 0$;
$x_{k+1} = x_k + \alpha_k d_k$; $k := k + 1$; *go to Step 2.*

The algorithm is actually a steepest descent method in the feasible region. Thus, its convergence can be established under certain line search conditions. When Z satisfies $Z^T Z = I$, Algorithm 11.5.1 is a projected gradient method.

Now we discuss a projected gradient method for general linearly constrained optimization problems, which was proposed by Calamai and Moré [50].

For a general linearly constrained optimization problem

$$\begin{aligned} \min_{x\in\Re^n} \quad & f(x), && (11.5.6)\\ \text{s.t.} \quad & a_i^T x = b_i, \quad i \in E, && (11.5.7)\\ & a_i^T x \geq b_i, \quad i \in I, && (11.5.8) \end{aligned}$$

the feasible set X is

$$X = \{x | a_i^T x = b_i,\ i \in E; a_i^T x \geq b_i,\ i \in I\}. \tag{11.5.9}$$

Define the mapping P,

$$P(x) = \arg\min\{\|z - x\|,\ z \in X\}, \tag{11.5.10}$$

where $\arg\min$ indicates any $z \in X$ that minimizes $\|z - x\|$, $\|\cdot\|$ is a norm. For simplicity, we assume that $\|\cdot\|$ is the Euclidean norm $\|\cdot\|_2$.

Consider the steepest descent method. The iterate x_{k+1} should be a point on the straight line

$$\bar{x}_k(\alpha) = x_k - \alpha \nabla f(x_k). \tag{11.5.11}$$

But we need the iterate points on the feasible region, we use the projection P to project the line (11.5.11) to X, obtaining the piecewise line

$$x_k(\alpha) = P[x_k - \alpha \nabla f(x_k)]. \tag{11.5.12}$$

We search along the piecewise line, namely find $\alpha_k > 0$ such that

$$f(x_k(\alpha_k)) \le f(x_k) + \mu_1 (x_k(\alpha_k) - x_k)^T \nabla f(x_k), \tag{11.5.13}$$

$$\alpha_k \ge \gamma_1 \quad \text{or} \quad \alpha_k \ge \gamma_2\ \bar{\alpha}_k > 0, \tag{11.5.14}$$

where $\bar{\alpha}_k$ satisfies

$$f(x_k(\bar{\alpha}_k)) > f(x_k) + \mu_2 (x_k(\bar{\alpha}_k) - x_k)^T \nabla f(x_k). \tag{11.5.15}$$

Here, $\gamma_1, \gamma_2, \mu_1, \mu_2$ are positive constants and $\mu_1, \mu_2 \in (0, 1)$.

The method of Calamai and Moré can be stated as follows.

Algorithm 11.5.2

Step 1. *Given a feasible point* x_1, $\mu \in (0,1)$, $\gamma > 0$, $\alpha_0 = 1$, $k := 1$;

Step 2. $\alpha_k := \max\{2\alpha_{k-1}, \gamma\}$.

Step 3. *if (11.5.13) holds go to Step 4;*
$\alpha_k = \alpha_k/4$; *go to Step 3;*

Step 4. $x_{k+1} := x_k(\alpha_k)$; $k := k+1$; *go to Step 2.*

It is easy to see that α_k computed by Algorithm 11.5.2 satisfies (11.5.13)-(11.5.15) for $\mu_2 = \mu_1 = \mu$, $\gamma_1 = \gamma$, $\gamma_2 = 1/4$.

From the definition of $P(x)$, for any $x \in \Re^n$ it follows that

$$(x - P(x))^T (z - P(x)) \le 0, \quad \forall z \in X. \tag{11.5.16}$$

Let $x = x_k - \alpha_k \nabla f(x_k)$ and $z = x_k$ in the above relation, then we obtain that

$$(x_k - \alpha_k \nabla f(x_k) - x_{k+1})^T (x_k - x_{k+1}) \le 0. \tag{11.5.17}$$

Thus, it follows from (11.5.13) and (11.5.17) that

$$f(x_k) - f(x_{k+1}) \geq \mu_1 \frac{\|x_{k+1} - x_k\|_2^2}{\alpha_k}. \tag{11.5.18}$$

First we have the following lemmas.

Lemma 11.5.3 *Assume that $f(x)$ is continuously differentiable and bounded below on the feasible set X. If $\nabla f(x)$ is uniformly continuous on X, the iterates generated by Algorithm 11.5.2 satisfy*

$$\lim_{k\to\infty} \frac{\|x_{k+1} - x_k\|}{\alpha_k} = 0. \tag{11.5.19}$$

Proof. If the lemma is not true, there is an infinite subsequence K_0 such that

$$\frac{\|x_{k+1} - x_k\|}{\alpha_k} \geq \delta, \quad \forall k \in K_0 \tag{11.5.20}$$

where $\delta > 0$ is a positive constant independent of k. It follows from the above relation and (11.5.18) that for all $k \in K_0$ we have that

$$f(x_k) - f(x_{k+1}) \geq \delta\mu_1 \|x_{k+1} - x_k\| \geq \delta^2 \mu_1 \alpha_k. \tag{11.5.21}$$

Because $f(x)$ is bounded below on the feasible set and all x_k are feasible, it follows that

$$\sum_{k=1}^{\infty} [f(x_k) - f(x_{k+1})] < +\infty. \tag{11.5.22}$$

Inequalities (11.5.21) and (11.5.22) imply that

$$\lim_{\substack{k\to\infty \\ k\in K_0}} \|x_{k+1} - x_k\| = \lim_{\substack{k\to\infty \\ k\in K_0}} \alpha_k = 0. \tag{11.5.23}$$

Therefore the first condition of (11.5.14) does not hold for sufficiently large $k \in K_0$, which shows that

$$\alpha_k \geq \gamma_2 \bar{\alpha}_k, \tag{11.5.24}$$

and that (11.5.15) holds. Using the monotonically non-increasing property of

$$\Psi(\alpha) = \frac{\|P(x + \alpha d) - x\|}{\alpha}, \quad \alpha > 0 \tag{11.5.25}$$

and relation (11.5.24), we can prove that

$$\frac{\|x_k - x_k(\bar{\alpha}_k)\|}{\bar{\alpha}_k} \geq \min\left\{1, \frac{1}{\gamma_2}\right\} \frac{\|x_k - x_k(\alpha_k)\|}{\alpha_k}. \tag{11.5.26}$$

Thus, letting $x = x_k - \bar{\alpha}_k \nabla f(x_k)$ and $z = x_k$ in (11.5.16) gives that

$$\begin{aligned} -(x_k(\bar{\alpha}_k) - x_k)^T \nabla f(x_k) &\geq \frac{\|x_k - x_k(\bar{\alpha}_k)\|^2}{\bar{\alpha}_k} \\ &\geq \min\left\{1, \frac{1}{\gamma_2}\right\} \delta \|x_k - x_k(\bar{\alpha}_k)\| \end{aligned} \tag{11.5.27}$$

for all sufficiently large $k \in K_0$. The uniform continuity of $\nabla f(x)$ on X implies that

$$f(x_k(\bar{\alpha}_k)) - f(x_k) = (x_k(\bar{\alpha}_k) - x_k)^T \nabla f(x_k) + o(\|x_k(\bar{\alpha}_k) - x_k\|). \tag{11.5.28}$$

It follows from (11.5.15) and (11.5.28) that

$$-(x_k(\bar{\alpha}_k) - x_k)^T \nabla f(x_k) \leq o(\|x_k - x_k(\bar{\alpha}_k)\|). \tag{11.5.29}$$

The above inequality contradicts (11.5.27), which shows that the lemma is true. $\square$

Lemma 11.5.4 *A point $x^* \in X$ is a KKT point of problem (11.5.6)-(11.5.8) if and only if there exists $\bar{\delta} > 0$ such that*

$$P(x^* - \alpha \nabla f(x^*)) = x^* \tag{11.5.30}$$

holds for all $\alpha \in [0, \bar{\delta}]$.

Proof. Equation (11.5.30) is equivalent to

$$\|x^* - \bar{\delta} \nabla f(x^*) - x^*\|_2^2 \leq \|x^* - \bar{\delta} \nabla f(x^*) - x\|_2^2 \tag{11.5.31}$$

holds for all $x \in X$. Because X is a convex set, (11.5.31) is equivalent to

$$(x - x^*) \nabla f(x^*) \geq 0 \tag{11.5.32}$$

holds for all feasible points sufficiently close to x^*. This means that x^* is the minimizer of function $x^T \nabla\ f(x^*)$ on X, which is equivalent to that x^* is a KKT point of problem (11.5.6)-(11.5.8). $\square$

From the above two lemmas, we can easily establish the convergence result of Algorithm 11.5.2.

Theorem 11.5.5 *Assume that $f(x)$ is continuously differentiable on the feasible set X. Then, any accumulation point x^* of $\{x_k\}$ generated by Algorithm 11.5.2 is a KKT point of problem (11.5.6)-(11.5.8).*

Proof. If the theorem is not true, there exist a subsequence of $\{x_k\}$ satisfying

$$\lim_{\substack{k\in K_0\\k\to\infty}} x_k = x^*, \tag{11.5.33}$$

and

$$P(x^* - \bar{\delta}\nabla f(x^*)) \neq x^*, \tag{11.5.34}$$

where $\bar{\delta} > 0$, K_0 is a subset of $\{1, 2, \cdots\}$. Because of (11.5.33), we can assume that $x_k \in S$ $(k \in K_0)$, and S is a bounded closed set. Because $\nabla f(x)$ is continuous on S, it is also uniformly continuous on S. It follows from Lemma 11.5.3 that

$$\lim_{\substack{k\in K_0\\k\to\infty}} \frac{\|x_{k+1} - x_k\|}{\alpha_k} = 0. \tag{11.5.35}$$

From the continuity of $\nabla f(x)$ and (11.5.33)-(11.5.34), we can show that

$$\lim_{\substack{k\in k_0\\k\to\infty}} \frac{\|x_k(\bar{\delta}) - x_k\|}{\bar{\delta}} = \frac{\|P(x^* - \bar{\delta}\nabla f(x^*)) - x^*\|}{\bar{\delta}} > 0. \tag{11.5.36}$$

Because the function $\Psi(\alpha)$ defined by (11.5.25) is monotonically non-increasing, it follows from (11.5.35) and (11.5.36) that $\alpha_k \geq \bar{\delta}$ holds for all sufficiently large $k \in K_0$. Therefore,

$$\begin{aligned} f(x_k) - f(x_{k+1}) &\geq -\mu_1(\nabla f(x_k))^T(x_k(\alpha_k) - x_k) \\ &\geq -\mu_1(\nabla f(x_k))^T(x_k(\bar{\delta}) - x_k) \\ &\geq \mu_1 \frac{\|x_k(\bar{\delta}) - x_k\|^2}{\bar{\delta}}. \end{aligned} \tag{11.5.37}$$

Now it follows from (11.5.37) and (11.5.36) that

$$\lim_{\substack{k\in K_0\\k\to\infty}} \inf[f(x_k) - f(x_{k+1})] > 0. \tag{11.5.38}$$

This contradicts the fact that $\lim_{k\to\infty} f(x_k) = f(x^*)$. Therefore the theorem is true. $\square$

Exercises

1. Assume that X is a convex polyhedral defined by $X = \{x \mid \quad Ax \geq b\}$. Show that finding a direction d satisfying (11.1.1)–(11.1.2) is a convex programming problem and give its dual.

2. By direct elimination, find the point on the ellipse defined by the intersection of the surface $x + y = 1$ and $x^2 + 2y^2 + z^2 = 1$ which is nearest to the origin.

3. Apply Newton's method with the generalized elimination to solve the problem

$$\begin{aligned} \min \quad & 8x_1^4 - x_2^4 \\ \text{s.t.} \quad & x_1 + x_2 = 1. \end{aligned}$$

4. For the above problem, at the point $(3, -2)^T$, please give the projected gradient and the projected Hessian. What are the projected gradient and the projected Hessian at the solution?

5. Give the projected gradient algorithm for the box constrained problem

$$\begin{aligned} \min \quad & f(x) \\ \text{s.t.} \quad & l \leq x \leq u. \end{aligned}$$

6. Prove Theorem 11.3.2.

7. Assume the symmetric matrix $B \in \Re^{n \times n}$ is invertible and $b \in \Re^n$. Prove that the matrix

$$\hat{B} = \begin{bmatrix} B & b \\ b^T & \beta \end{bmatrix} \tag{11.5.39}$$

is invertible if and only if $\beta - b^T B^{-1} b \neq 0$. And prove that, when $\hat{B}$ is invertible, there exist μ and u such that

$$\hat{B}^{-1} = \begin{bmatrix} B^{-1} & 0 \\ 0 & 0 \end{bmatrix} + \mu u u^T. \tag{11.5.40}$$